What makes motion hard to stop?

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Learning target: Compare momentum changes in a controlled stopping situation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 14.*
Fastest at 3 AU. Sweeping equal areas in equal times requires greater speed near the Sun.

02. Worked example: explain the model
Velocity arrows equal; momentum arrow for 4 kg cart twice as long.

03. Read and interpret the model
B: 6 kg·m/s versus A: 2 kg·m/s; velocity is held fixed.

04. Apply the physics
A: initial momentum +2 kg·m/s, Δp = -2 kg·m/s, average force = -2/3 N. B: initial momentum +6 kg·m/s, Δp = -6 kg·m/s, average force = -2 N. Both forces point left.

05. Explain your reasoning
No. Stopping time, energy, deformation, contact area and material response matter.

06. Review: connect an earlier idea *Review: Unit 2, Day 5.*
4 N, attractive toward the other mass.

07. Exit check
Momentum-change magnitude = 6 kg·m/s. Average force magnitude = 6/3 = 2 N, directed left.

08. STRETCH: test the model
Example 2 kg at 6 m/s gives 12 kg·m/s right.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 14.*
$T_B/T_A=3^{(3/2)}\approx5.196$. The normalization depends on the central mass and chosen units.

02. Worked example: explain the model
Velocity arrows equal; momentum arrow for 4 kg cart twice as long.

03. Read and interpret the model
B: 6 kg·m/s versus A: 2 kg·m/s; velocity is held fixed.

04. Apply the physics
Δp=-2,-6,-4 kg·m/s; forces=-4,-12,-8 N.

05. Explain your reasoning
No. Stopping time, energy, deformation, contact area and material response matter.

06. Review: connect an earlier idea *Review: Unit 2, Day 5.*
$F\approx2.964\times10^{-5}$ N, attractive. The incorrect numerical answer is 6 times too large when SI values are used.

07. Exit check
$J = 3(0 - 4) = -12$ N·s; $F_{avg} = -12/4 = -3$ N.

08. STRETCH: test the model
Each momentum +4 kg·m/s; change -4; average force -20 N. Equal momentum change over equal time gives equal average force.